StudentProfile AI

Mini Project – Multi-Step Prompt Workflow for Student Profile Understanding

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PROJECT NAME	StudentProfile AI
PROJECT TYPE	Mini Project – Prompt Workflow Design

Objective: Understand a student's academic, technical and personal learning profile using a structured AI prompt chain.

01 • Project Objective

The objective of StudentProfile AI is to design a structured, multi-step prompt workflow that takes a student's academic information, skills, projects, interests and learning goals as input. The system identifies the student's current profile, organizes strengths and areas for improvement, recommends suitable learning priorities, creates a personalized development roadmap, and produces a practical student action plan. The workflow demonstrates prompt chaining, where the output of one stage is deliberately passed as context to the next stage.

02 • Problem Statement

Students often have information about their education, skills and projects scattered across resumes, marksheets, project descriptions and personal goals. It can be difficult to understand their overall profile, identify strengths, recognize weak areas, decide what to learn next and plan suitable projects. A single AI prompt may give general advice, but it may not provide a systematic and traceable understanding of the student. StudentProfile AI solves this by dividing the task into connected analysis stages.

03 • Workflow Architecture

INPUT	STUDENT DATA	PROFILE UNDERSTANDING	SKILL & ACADEMIC ANALYSIS	STRENGTHS & GAPS	PRIORITY & SCORING	LEARNING ROADMAP	STUDENT ACTION PLAN

04 • Workflow Steps & Expected Outputs

1. Student Profile Understanding	Extract education, academic performance, technical skills, programming languages, tools, projects, certificatio	Structured student profile
2. Academic & Skill Analysis	Analyze subjects, grades, technical abilities, soft skills and project experience	Academic and skill map
3. Strength & Gap Detection	Identify demonstrated strengths and areas needing improvement using evidence	Strength-gap matrix
4. Priority & Scoring	Score improvement areas based on goal relevance, proficiency and estimated learning effort.	Prioritized scoring table
5. Personalized Learning Roadmap	Convert priorities into an ordered learning plan with practice, projects and milestones	Learning roadmap
6. Student Action Plan	Combine findings into project, resume, portfolio, interview and immediate actions.	Actionable development plan

05 • Prompt-by-Prompt Workflow

01 • Student Profile Understanding

Prompt: Analyze the provided student information. Extract education, academic performance, technical skills, programming languages, frameworks, tools, projects, certifications, interests, extracurricular activities and learning goals. Distinguish explicitly stated information from assumptions. Return the result using clear headings and concise bullet points. Do not recommend anything yet.

Expected Output: Structured student profile with academic context, skill evidence, interests and goals.

02 • Academic & Skill Analysis

Prompt: Using the student profile, analyze academic subjects, technical skills, soft skills, project experience and areas of demonstrated competence. Group related skills and distinguish strong, moderate and developing areas. Do not create a learning plan yet.

Expected Output: Academic and competency map with strengths and development areas.

03 • Strength & Gap Detection

Prompt: Using the student profile and academic-skill map from the previous stages, identify major strengths and improvement gaps. For every gap, cite the relevant evidence available in the input context. Do not invent achievements, skills or experience.

Expected Output: Strength-gap matrix with status, evidence and concise explanations.

04 • Priority & Scoring

Prompt: Using the strength-gap matrix, prioritize development areas based on relevance to the student's stated goals, current proficiency and estimated learning effort. Give each important area a score from 1–10 and label it High, Medium or Low priority. Explain the score briefly and identify the top three areas that would most improve readiness.

Expected Output: Prioritized development scorecard and top-three priorities.

05 • Personalized Learning Roadmap

Prompt: Using the prioritized development areas, create an ordered learning roadmap. Organize it into realistic phases or weeks, with a learning objective, practice task, small project/application idea and measurable completion criterion for each phase. Build prerequisites into the order and avoid recommending skills already demonstrated unless they need strengthening.

Expected Output: Personalized roadmap with milestones, practice and project work.

06 • Student Action Plan

Prompt: Using all previous outputs, create a concise student action plan. Include recommended resume/portfolio improvements, project actions, interview preparation topics, skills to demonstrate and immediate next steps. Clearly distinguish evidence-based recommendations from optional suggestions. Do not invent achievements, certifications or experience.

Expected Output: Final student-profile report containing strengths, gaps, roadmap and actionable next steps.

06 • Prompt Chaining, Demonstration & Learning Outcome

The workflow is designed so that each stage builds on structured information produced earlier. Step 1 creates the student profile; Step 2 analyzes academic and skill information; Step 3 identifies strengths and gaps; Step 4 prioritizes those gaps; Step 5 transforms priorities into a learning roadmap; and Step 6 consolidates the findings into a student action plan. This makes the workflow reproducible, traceable and more useful than a single broad student-advice prompt.

STAGE	CONTEXT PASSED FORWARD	DECISION ENABLED
1 → 2	Student profile	Understand academic and skill context.
2 → 3	Student profile + academic/skill map	Detect strengths and improvement gaps.
3 → 4	Strength-gap matrix	Determine which development areas matter most.
4 → 5	Prioritized areas + scores	Build learning order and milestones.
5 → 6	Learning roadmap + previous findings	Create a complete student action plan.

07 • Demonstration Example

Input Student: A final-year student studying Computer Science with Java, SQL, React and Node.js experience, two academic projects and basic Git knowledge. The student wants to improve backend development skills and become more job-ready.

Student Goal: Become prepared for junior software development roles with stronger backend and project-development skills.

STAGE	EXAMPLE RESULT
Profile Understanding	Current strengths: Java, SQL, React, Node.js and basic Git. Evidence: academic projects using Java and web technologies.
Academic & Skill Analysis	Strong: Java and web development. Developing: backend architecture, API development and production deployment.
Strength & Gap Detection	Strengths: Java, SQL and project experience. Gaps: Spring Boot, REST API development, PostgreSQL depth and deployme
Priority	1. Spring Boot – High; 2. REST APIs – High; 3. PostgreSQL – Medium; 4. Deployment/cloud basics – Medium.
Roadmap	Phase 1: Spring Boot basics → Phase 2: REST API + PostgreSQL project → Phase 3: deployment and interview practice.
Action Plan	Build one backend project, document it in a portfolio/GitHub repository, strengthen interview topics and demonstrate the proje

08 • Expected Learning Outcome

This project demonstrates how prompt engineering can decompose student-profile understanding into manageable AI interactions. It highlights sequential prompting, contextual reuse, structured outputs, comparison, scoring, prioritization, iterative planning and final action generation. The approach encourages evidence-based recommendations rather than unsupported claims.

09 • Conclusion

StudentProfile AI transforms student information into a structured understanding of academic background, skills, strengths, development gaps, learning priorities and next actions through six connected prompt stages. By separating profile extraction, academic and skill analysis, gap detection, prioritization, roadmap generation and action planning, the project provides a clear and practical approach to AI-assisted student development.

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